RISKSIGNAL

# The M2 student’s presentation guide

Operational Risk foundations, the maths behind the prototype, what was built, what to say honestly, a focused 20-minute script, and a short email teaser.

**The one-sentence project**
RiskSignal is a research prototype that helps an analyst move from a technology-risk signal to a traceable investigation by connecting synthetic incidents, indicators, controls, dependencies and transparent quantitative tools.

**Audience:** Operational Risk managers and reviewers
**Author position:** M2 Applied Mathematics / IASD student, learning the OR domain
**Important:** This guide uses plain language. It is not a claim that the prototype is a validated bank model.

Prepared from the RiskSignal repository and the supplied CV. Project facts checked against code and documentation.

# Start here: what you must be able to explain

If you remember only these points, you can still present the project responsibly.

- Operational Risk concerns loss or disruption caused by people, processes, systems or external events. This prototype focuses on technology-related examples.
- Atlas Bank is fictional. Its records and losses are synthetic; the PAYMENT_ENGINE_01 storyline was deliberately designed for a readable demonstration.
- The tool connects observed records. A shared supplier, rising indicator or overdue review is a prompt for investigation, not proof of cause or control failure.
- The quantitative model is a compound Poisson frequency-severity illustration: a random number of incidents, random positive losses, summed into hypothetical annual totals.
- Monte Carlo outputs describe what follows from the chosen assumptions. They are not forecasts, expected real bank losses or regulatory capital.
- The LLM does not calculate the loss model. It can select approved read-only tools and explain returned evidence. The offline mock is a deterministic script, not a live LLM.
- Passing software tests checks code behaviour and calculations. It does not validate the assumptions or show real-world predictive performance.
- Your credible contribution is the problem framing, mathematics, choices, tests and limitations you can defend—not a claim that you typed every line yourself.

> **Safe opening**
> “I am presenting an AI-assisted research prototype. It uses a fictional bank and illustrative assumptions. The goal is to demonstrate a traceable investigation workflow, not to predict real losses or make a bank decision.”

## Quick contents


1. Operational Risk in plain English
2. Atlas Bank and what the app does
3. The maths, results and interpretation
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5. Your M2 fit and the honest development story
6. Credibility checklist and manager questions
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# 1. Operational Risk, without assuming prior OR knowledge

A useful way to approach a new domain is to define the work before explaining the software.

**Operational Risk (OR)** is the risk of loss or disruption because internal processes, people or systems fail, or because an external event affects the organisation. Examples include a failed technology release, a service outage, an error in a process, or a supplier disruption. RiskSignal narrows this broad subject to technology and third-party dependencies.

A bank is a connected service, not just a set of applications. A customer payment may rely on a payment engine, a messaging service, a vendor connection, a control that checks releases, and a business process that completes settlement. An analyst asks what happened, what else may be affected, which records support that view, and what should be checked next.

| Term | Plain-English meaning | How to discuss it in RiskSignal |
|---|---|---|
| System | Technology used to deliver a service. | The focal example is PAYMENT_ENGINE_01. |
| Business process | Work the organisation performs, such as sending a payment. | A process relies on one or more systems. |
| Incident | A recorded event that went wrong or disrupted work. | Synthetic history supplies event counts and loss observations. |
| Loss | The financial amount attributed to an event. | Synthetic units only; no real currency is assigned. |
| Near miss | An event that could have caused loss but did not. | Useful context; it is not included in the frequency-severity model. |
| Control | A safeguard intended to prevent a problem or limit its effect. | A review date or overdue label is evidence to check, not proof it works or failed. |
| KRI | Key Risk Indicator: a measure used to watch a risk condition. | A change can prompt review; here it is not a predictor or approved appetite limit. |
| Vendor / third party | An outside provider a system depends on. | A vendor shared by systems can create a shared exposure. |
| Dependency | A relationship describing what needs what to work. | A graph helps find linked systems, processes and suppliers. |
| RCSA | Risk and Control Self-Assessment: a structured way to identify risks and assess controls. | This app prepares evidence and questions; it does not produce a valid control rating. |
| Scenario | A structured “what if?” with stated assumptions. | The shared-vendor stress changes model inputs; it is not a prediction. |

**Distinction to repeat**

An observation is not automatically an explanation. “The control review is overdue” is an observation. “The control failed and caused the incident” is a causal conclusion that these data cannot support.

# 2. What the prototype does

The core journey is: notice a relevant signal, ask the agent to investigate, review evidence, then decide what to examine next.

| Step | What the analyst sees or does | Why it matters |
|---|---|---|
| 1. Select | Choose a technology system. | Keeps the investigation focused. |
| 2. Observe | See up to four prioritised monitoring indicators. | Provides a small amount of context without opening a wall of charts. |
| 3. Ask | Enter a question or choose Investigate System. | The human sets the investigation goal. |
| 4. Retrieve | The agent calls read-only evidence tools. | Incidents, KRIs, controls, dependencies and optional simulations are fetched by code. |
| 5. Review | Read findings first; expand source records and tool activity when needed. | Progressive disclosure keeps supporting detail available and traceable. |
| 6. Decide | The analyst chooses whether to request follow-up. | No model or agent makes the final risk decision. |

## What is in Atlas Bank?

The default seeded dataset contains 20 systems, 10 business processes, 8 suppliers, 40 controls, 312 incidents and about 5,880 monthly KRI observations. The observation period is 1 October 2021 through 30 September 2026. The generator validates references and values and writes CSV tables and a SQLite database. Seed 42 makes the generated dataset reproducible.

Some elements are hand-designed and some use random draws. In particular, PAYMENT_ENGINE_01 has deliberately placed recent incidents, an overdue regression-testing review, drifting indicators and a shared vendor. Other systems have example signals too. This is a designed teaching case, not an outcome discovered by analysis.

## The navigation

**Introduction** describes the objective and shows a small sample. **System Explorer** provides detailed records and dependency relationships. **Scenario Analysis** exposes the assumptions and simulated comparisons. **Risk Investigator** puts the compact monitoring context and evidence-led agent in one workspace. **Way forward** explains limits and proposed next steps.

> **Do not say**
> “The system discovered a real risk pattern.” Say instead: “The system connects a deliberately constructed example so we can demonstrate the investigation workflow.”

# 3. The mathematical model you need to understand

You do not need to memorise every formula. You should be able to explain what each assumption means and where it could fail.

## 3.1 Frequency: how many incidents?

Let N be the number of incidents in a year. The prototype assumes N follows a Poisson distribution with rate lambda. Lambda is the average count per year. For PAYMENT_ENGINE_01, 19 synthetic incidents over approximately 5 years give lambda about 19 / 5 = 3.8 per year.

**Assumption:** incidents arrive independently at a steady average rate. Real operational events may cluster, change over time, or become more frequent after a common cause. A Poisson model may then be too simple. The system uses a bank-wide per-system fallback when one system has too few counts; the selected method is reported.

## 3.2 Severity: how large is each loss?

Let X be the loss from one incident. The prototype uses a Lognormal distribution: ln(X) is Normal with parameters mu and sigma. This keeps losses positive and allows a long right tail. Mu and sigma are parameters in log space. exp(mu) is the median loss; it is not the average loss. Sigma controls spread in log space.

For a Lognormal, the average single-event loss is exp(mu + sigma squared / 2). The prototype fits these parameters using observed positive losses, choosing the most specific data set with enough records. The focal system has 19 synthetic loss records. Because the generator itself creates losses from Lognormal distributions, fitting a Lognormal back to them is circular and is not evidence that real bank losses follow that distribution.

## 3.3 Annual aggregate loss: combine the two

For each imaginary year, draw N from the Poisson distribution, draw N individual losses from the Lognormal distribution, then add them. If N is zero, that year's loss is zero. This is a compound Poisson frequency-severity model. It separates how often an event occurs from how costly it is.

## 3.4 Monte Carlo: play out many modelled years

The annual-total distribution is generated by simulation. RiskSignal draws 100,000 hypothetical years by default, using a fixed random seed. Each year is generated from the same stated assumptions. A seed makes the software output repeatable; it does not make the assumptions true.

## 3.5 Read the tail measures

| Measure | Say it simply |
|---|---|
| Mean | Average of the simulated annual totals; a few large years can pull it upward. |
| Median | Middle simulated year: half of the totals are lower, half higher. |
| P95 / VaR 95 | The modelled annual-loss threshold exceeded by about 5% of simulated years. |
| P99 / VaR 99 | The threshold exceeded by about 1% of simulated years. |
| ES95 | Average annual total among the worst 5% of simulated years. |
| ES99 | Average annual total among the worst 1% of simulated years. |

Expected Shortfall is a tail average, not just a threshold. P99 and ES99 rely on fewer simulated years than the mean, so they are noisier. More simulation reduces random simulation noise, but does not fix poor data, uncertain parameters or a wrong model.

# 4. The example numbers and what they do not mean

These default values are useful for rehearsing the interface. Read the current screen during the presentation if its settings differ.

| Default example | Value | Careful interpretation |
|---|---|---|
| Observed incidents | 19 over about 5 years | Synthetic history for one system. |
| Frequency rate | lambda about 3.80 / year | A fitted average under the Poisson assumption. |
| Lognormal parameters | mu 10.114; sigma 1.478 | Fit to 19 synthetic positive losses; both are log-space parameters. |
| Simulated mean annual loss | 278,433 | Synthetic monetary units; conditional on assumptions. |
| Median annual loss | 163,846 | Typical simulated year under the chosen model. |
| P95 / ES95 | 896,435 / 1,546,928 | Tail measures from simulation, not a forecast or capital figure. |
| Shared-vendor stress P95 | 2,276,896; about +154% | Output after applying a pre-defined illustrative scenario. |

## How to explain the stress

The SHARED_VENDOR_OUTAGE scenario assumes a frequency multiplier of 1.8, adds 0.2 to mu, and multiplies sigma by 1.1. These are explicit hypothetical choices, not estimates of a real outage. The engine uses common random numbers for baseline and stress so corresponding simulated years share underlying random draws, reducing simulation noise in their comparison. That careful numerical comparison does not validate the scenario assumptions.

> **Safe interpretation**
> “Under these synthetic data and stated assumptions, the simulation gives this P95. It is an illustrative model output, not the bank's expected loss, a forecast, a capital number, or a calibrated estimate.”

## Why not build a predictor?

The generator has no hidden future-incident label and no planned link between KRI values and later incidents. A model trained on this synthetic data could learn artifacts of the generator, not general bank behaviour. The current project therefore demonstrates descriptive evidence and stochastic mechanics; it does not claim machine-learning prediction.

# 5. Monitoring, controls and dependency reasoning

The operational evidence is useful because it creates questions. It is not proof by itself.

## Monitoring indicators

The workspace shows at most four selected KRI cues. The supporting monitoring method compares an observation with prior values using a trailing median and median absolute deviation (MAD), with explicit handling for missing, stale, gapped and flat histories. A one-sided unusual-increase rule is an illustrative review policy. It is not an approved risk-appetite threshold and has not been calibrated against real incidents. Flag counts describe review workload, not model precision or recall.

## Controls and RCSA prompts

A control is a safeguard; a review date tells us whether a scheduled review is due or overdue. RiskSignal can surface a record and suggest a question, but it does not have evidence that the control operated effectively. Its RCSA section is evidence preparation, not a complete risk assessment, control test, rating or approval process.

## Dependencies and shared exposure

The dependency graph represents directed links among systems, business processes and suppliers. If PAYMENT_ENGINE_01 and PAYMENT_ENGINE_02 both depend on V04 Keystone Payment Rails, V04 is a shared exposure: a reason to consider whether both systems are relevant to an investigation. The relationship alone cannot prove that a vendor caused an incident, or that one system's outage will cause another's. The loss simulation is still run for one selected system at a time; it does not model joint losses spreading across the graph.

> **Useful sentence for a manager**
> “The graph tells us which records are connected. It does not estimate the probability or financial impact of a supplier failure propagating through the whole bank.”

# 6. What the AI agent actually does

The important design split is: Python retrieves and calculates; an LLM may choose tools and write a summary.

| Component | Responsibility |
|---|---|
| Data and SQLite | Store validated fictional systems, incidents, controls, dependencies and KRIs. |
| NetworkX graph | Answer relationship and shared-dependency questions. |
| NumPy / pandas model | Estimate illustrative parameters and simulate annual losses. |
| Typed read-only tools | Expose a narrow set of retrieval and analysis functions. |
| Agent loop | Send the question to the selected provider, execute requested tools, return tool results, repeat within a turn limit, and record activity. |
| FastAPI | Expose typed backend endpoints. |
| Streamlit / Plotly | Present selection, compact risk context, investigation findings and expandable evidence. |

## Live LLM versus offline mock

**Live mode:** a configured Anthropic provider can use tool calling. The model's tool selection and prose are generated live. The live path was not exercised in development because no key was available; it must be tested separately before a live presentation.
**Mock mode:** a deterministic script follows predefined responses. It is repeatable and useful for tests and an offline demo, but it is not a live language model. Never call it a live AI result.

## What is and is not enforced

- The model has no direct database access. It can use only the registered tools.
- The registry checks tool names and arguments. Tools are read-only; there is no action to edit or close risk records.
- Numbers are retrieved from tools, and the output is scanned for numbers that do not match tool results. This is a best-effort check, not semantic proof.
- A right number can still be attributed to the wrong system or explained with an unsupported cause. Human review remains necessary.
- Tool activity and result payloads support traceability. The model's internal reasoning is not recorded.

> **Say this precisely**
> “The agent is an evidence assistant, not a risk decision-maker. The mock provider is scripted. I distinguish it from live LLM tool calling.”

# 7. How this connects to your M2 IASD

Your mathematics and data-science training gives you tools for building and checking the prototype. Operational Risk vocabulary and governance are the domain you are learning.

| IASD / CV evidence | Connection to RiskSignal | Honest boundary |
|---|---|---|
| Probability, Bayesian statistics, stochastic calculus and Poisson processes | Understand random event counts, uncertain outcomes and stochastic modelling choices. | Course knowledge does not make the chosen model suitable for a real bank. |
| Monte Carlo methods and options pricing | Understand simulation, seeds, tail summaries and sampling noise. | A Monte Carlo result is only as relevant as its input assumptions. |
| High-dimensional statistics and optimisation | Bring quantitative reasoning and disciplined parameter thinking. | This prototype is not a high-dimensional predictive model. |
| Python numerical modelling internship; finite-volume solver; Sobol sampling across parameters | Transfer experience building reusable numerical workflows and exploring parameter variation. | Energy-system modelling is not Operational Risk expertise; explain the transfer, not equivalence. |
| Mathematical proof/research experience | Support careful assumptions, definitions, reasoning and checking edge cases. | Domain interpretation still needs review from OR practitioners. |

## The process of creating this project

**1. Frame the question.** Explore the idea: help an analyst connect a technology-risk signal to relevant evidence and possible next questions. This is a proposed workflow, not a proven efficiency gain.

**2. Set boundaries.** Use fictional data; keep controls and dependencies as evidence; do not claim prediction, causal discovery, real-bank calibration or automated decisions.

**3. Build a reproducible data foundation.** Define systems, processes, vendors, controls, incidents and KRIs; generate and validate a seeded dataset; save CSV and SQLite forms.

**4. Implement the analytical core.** Build frequency and severity estimators, a seeded aggregate loss simulator, tail metrics, stress comparisons and dependency queries as separate Python functions.

**5. Expose bounded tools and an agent.** Put retrieval/calculation behind typed read-only functions. Let an LLM provider request a tool; let Python run it; return results for synthesis. Keep a script-based mock provider for repeatable offline tests.

**6. Connect the API and interface.** FastAPI exposes the backend; Streamlit calls it through an HTTP client. The investigator page shows only a few signals first, then lets the user expand evidence.

**7. Check the implementation.** Use unit, integration and UI tests, deterministic seeds, independent numerical checks and clear error paths. A passing test suite checks software behaviour, not empirical validity.

**8. Review and improve.** Check the design with domain experts, improve language, remove unhelpful claims, and disclose limitations. Development is AI-assisted: be transparent that an AI coding assistant helped implement the software; own the problem definition, choices, verification and ability to explain it.

# 8. Tools used and how to describe your role

| Tool / technology | What it does here |
|---|---|
| Python | Main implementation language and application integration. |
| NumPy / pandas | Random simulation, arrays and structured incident/KRI data. |
| SciPy | Statistical utilities used in quantitative checks and parts of the project. |
| NetworkX | Builds and queries the dependency graph. |
| Pydantic | Typed records and API/tool input-output validation. |
| SQLite | Local structured storage for the synthetic dataset. |
| FastAPI | HTTP API between the interface and backend functions. |
| Streamlit / Plotly | Analyst-facing application and charts. |
| Anthropic API (optional) | Live LLM provider when configured; not exercised in development. |
| Mock provider | Deterministic scripted replacement for offline tests/demo; not a live LLM. |
| pytest | Automated tests; the verified suite for this working tree passed 691 tests. |
| Docker Compose | Packaging path exists, but Docker images were not runtime-verified in development. |

## How to answer “Did you code all of it?”

> *““I originated and scoped the project idea and used an AI coding assistant to help implement much of the code. I do not want to overstate manual authorship. My responsibility is to understand the design, review the decisions, run and inspect the tests, and explain what the system can and cannot claim. I can show the architecture and the reasoning behind the main modelling choices.””*

This answer is stronger than pretending you wrote every line. Before presenting, rehearse explaining one data flow end to end, one formula, one test, one limitation and one next step without reading.

**Do not claim**

Do not say you independently designed or validated a bank-grade OR model, trained an accurate risk predictor, discovered real causal relationships, measured time saved, or production-hardened the system.

# 9. Credibility checklist and questions managers may ask

A credible M2 presentation is technically precise, clear about uncertainty, and comfortable saying “not yet.”

## Before the meeting

- Run the application and tests in advance. Use the offline mock if the live provider has not been rehearsed.
- Know which provider is active and say it accurately. The mock is a script; live provider requires a working key and model.
- Read the generated data seed and scenario assumptions; do not quote values from memory if the screen differs.
- Prepare one example of an observation, one source record that supports it, and one question the analyst should ask.
- Be ready to explain Poisson rate, Lognormal parameters, P95 and ES95 in plain language.
- Say synthetic, illustrative, not calibrated, not independently validated, and not a forecast before someone asks.
- Keep the app local: it has no authentication and is not suitable for confidential bank data.

| Question | Credible answer |
|---|---|
| Is this a prediction? | No. There is no future-loss target or predictive validation. It demonstrates retrieval, descriptive evidence and a model under stated assumptions. |
| Are the numbers real or in dollars? | No. The data are generated for a fictional bank and amounts are synthetic units, not a specified currency. |
| What does the AI add? | In live mode, it can choose from narrow read-only tools and organise their results into a summary. Calculations remain in Python. The mock version is scripted. |
| Can the agent decide risk severity or remediation? | No. It can surface evidence and suggest review questions. A human assesses context and decides follow-up. |
| Does an overdue control mean control failure? | No. It means the scheduled review date has passed. A review date is not a control operating-effectiveness test. |
| Does a shared vendor mean it caused both incidents? | No. It indicates shared dependency and concentration exposure. It does not establish cause or joint loss probability. |
| Why Poisson and Lognormal? | They provide a transparent introductory frequency-severity model. Counts and positive right-skewed losses fit the mechanics; suitability for a bank is not established. |
| Is the model validated? | No. Tests and independent numerical checks support implementation correctness. Real-data calibration, backtesting and independent validation are not done. |
| Why not use machine learning? | The synthetic generator contains no suitable future outcome label. Training a predictor here would risk learning how the data were generated. |
| What would you do next? | Agree one real analyst task and approved dataset with a risk team, compare the prototype with current practice, review evidence quality and errors, and measure usefulness, time, cost and latency before any wider pilot. |

# 10. Twenty-minute presentation: the run of show

Aim for about 17-18 minutes of prepared material, leaving a short buffer for questions or a slow page load. If there are no questions, use the closing and next-step section rather than adding technical detail.

| Time | Part | What to achieve |
|---|---|---|
| 0:00-1:00 | Opening | State the prototype, synthetic data and boundary. |
| 1:00-3:00 | OR problem and project question | Define OR and frame a question, not an unsupported industry claim. |
| 3:00-5:00 | Data and case | Show Atlas Bank and explain the deliberately designed focal example. |
| 5:00-7:00 | How the application works | Explain evidence flow and separation of duties. |
| 7:00-10:00 | System Explorer | Show one incident, one control record and one shared dependency. |
| 10:00-13:00 | Scenario Analysis | Explain the model and one conditional result. |
| 13:00-16:00 | Risk Investigator | Run the workflow; findings first, source evidence second. |
| 16:00-18:00 | Limits, next step and M2 fit | Demonstrate technical maturity and honest self-positioning. |
| 18:00-20:00 | Questions / close | Answer briefly or give the planned concise conclusion. |

## 0:00-1:00 | Opening

**Screen:** Introduction page. Pause on the project title and synthetic-data notice.

> *““Good morning. RiskSignal is an AI-assisted research prototype for technology Operational Risk. It connects incidents, monitoring indicators, controls and system dependencies in one investigation workflow. Everything I will show is synthetic. The loss model is illustrative, and this is not a forecast or a production decision tool.””*

## 1:00-3:00 | What OR means and the question

**Screen:** Introduction objective and workflow.

> *““Operational Risk is the possibility of loss or disruption from failures in people, processes, systems or external events. My focus here is technology systems and their suppliers. I am not claiming that every bank team works in the same way or that current work is slow. My design question is narrower: if an analyst starts from one technology signal, can a single workspace help bring relevant records together and make the next review question clearer?””*

## 3:00-5:00 | Atlas Bank and the data

**Screen:** Atlas Bank sample; point to the system profile and dependency visual.

> *““Atlas Bank is fictional and reproducible. The default dataset includes 20 systems, 10 business processes, 8 vendors, 40 controls, 312 incidents and around 5,880 monthly indicator observations across five years. Seed 42 reproduces the same data. The payment-system example has deliberately placed recent incidents, an overdue review and a shared supplier. That is a demonstration design, not a discovery about a real bank.””*

## 5:00-7:00 | How the system works

> *““The data sit in SQLite. Python functions handle the dependency graph and the loss simulation. Typed read-only tools expose those functions. FastAPI serves them to the Streamlit interface. In live mode, a language model may choose tools and write a summary; the calculations remain deterministic Python. The model has no write tool, so it cannot change risk records.””*

Transition: “I will now follow one system through the evidence, the quantitative illustration and the investigation workspace.”

## 7:00-10:00 | System Explorer: observe, do not over-interpret

**Screen:** Select PAYMENT_ENGINE_01. Show one incident, C002, and the shared V04 relationship.

> *""Here is a recent synthetic incident. The record tells us an event was logged; it does not by itself explain the cause. The control C002, Pre-Release Regression Testing, is overdue for review. That is a prompt to check current test evidence, not proof that the control failed or caused an incident.""*

> *""The graph shows that the payment engine and another system share vendor V04. This is a shared exposure and a reason to ask whether both systems matter to the investigation. The link does not prove a causal effect, and the loss simulation does not calculate a bank-wide outage spreading across the graph.""*

If showing a KRI, say: "This is an illustrative past-only review signal. It is not an approved appetite threshold and it does not predict an incident."

## 10:00-13:00 | Scenario Analysis: model the assumption, not the bank

**Screen:** Show the baseline parameters and the predefined shared-vendor scenario.

> *""The frequency estimate is about 3.8 incidents per year, based on 19 synthetic incidents over about five years. The severity model is Lognormal: losses are positive and can have a long right tail. Mu and sigma describe the log losses, not the loss mean and standard deviation. The generator itself created Lognormal losses, so fitting a Lognormal back is not evidence of real-world fit.""*

> *""Monte Carlo simulation plays out 100,000 hypothetical years: draw a count, draw losses and add them. P95 is the modelled threshold exceeded in about 5 percent of those years. Expected Shortfall 95 is the average of the worst 5 percent. Under these assumptions, the baseline P95 is about 896 thousand synthetic units. This is not a forecast, not USD, and not capital.""*

> *""The stress scenario increases the frequency assumption, shifts the typical severity and widens the log-loss spread. The P95 rises to about 2.28 million, around 154 percent. That is what these chosen assumptions produce. It is not evidence that a real vendor outage would have this effect.""*

## 13:00-16:00 | Risk Investigator: the main workflow

**Screen:** Risk Investigator; select the system if needed; click Investigate system.

> *""The page starts with a few relevant monitoring cues and passes them as context. The investigation can connect them with incidents, controls, dependencies and, when requested, quantitative results. I read the summary first, then expand the evidence to check the source behind a statement.""*

> *""The provider matters. In mock mode the response is scripted and repeatable; it is not a live LLM. In live mode a configured LLM can request the read-only tools. In both cases the analyst must check the evidence and make the decision.""*

Action: open one evidence item and point to the tool activity. Say: "This trace shows which tool was called and the result returned. The numeric check is helpful, but it cannot prove the prose has the right causal interpretation."

## 16:00-18:00 | Limits, next step and your M2

*““The main limits are synthetic records, illustrative parameters, no predictive target, no real-data validation, no authentication, and no persistent case audit service. The next sensible step is not to add more AI. It is to agree one analyst question and approved data with a risk team, build reviewed reference answers, and compare the evidence, errors and usefulness with today's process.””*

*““My M2 IASD contributes probability, statistics, stochastic modelling, Monte Carlo and Python implementation. My quantitative-modelling internship also involved building a reusable numerical workflow and exploring many parameter combinations. Operational Risk is the domain I am learning; I would expect practitioners to challenge the definitions, controls and scenario assumptions.””*

## 18:00-20:00 | Close or questions

*““The project demonstrates a bounded path from a signal to evidence and suggested review questions. It does not demonstrate that an AI can predict operational losses. Its value for me is combining mathematical modelling, software engineering and careful evidence communication while making the assumptions visible. Thank you. I would welcome your questions, particularly on which analyst task would be most useful to test first.””*

If no questions: use the remaining time to repeat the proposed pilot test and the mock/live distinction. Do not add extra equations.

## Short answers to likely follow-up questions

| If asked | Answer in one sentence |
|---|---|
| What is the main contribution? | A connected, evidence-led investigation workflow with deterministic tools and explicit limits. |
| What is the biggest model risk? | The small synthetic sample and assumed distributions; parameter uncertainty and model uncertainty dominate what the output can mean. |
| What would make it credible in a real setting? | Approved data, expert-defined scenarios and thresholds, independent review, out-of-time evaluation, security and a measured analyst trial. |
| Why use a mock? | To test the tool and UI workflow reproducibly without a model key; it is explicitly not presented as live LLM behaviour. |

# 11. Email teaser: share the idea, not the whole project

Do not attach this full guide or send a large code bundle in the first message. Offer a short conversation and a one-page overview only if requested.

## French version

**Objet :** Prototype IASD - investigation de risques opérationnels technologiques

*“Bonjour [Nom],<br/><br/>Je termine un prototype de recherche réalisé dans le cadre de mon parcours en mathématiques appliquées et IASD. RiskSignal explore une question ciblée : comment aider un analyste à relier des incidents, indicateurs, contrôles et dépendances fournisseurs avant de décider quoi examiner ensuite ? La démonstration utilise uniquement des données synthétiques et un modèle de pertes illustratif ; elle ne prétend ni prédire les pertes ni évaluer un risque bancaire réel.<br/><br/>Le projet combine Python, simulation stochastique, graphe de dépendances, API et assistant à outils en lecture seule. Je serais heureux d'avoir votre avis sur le cas d'usage et sur les limites à traiter avant tout essai avec des données approuvées. Auriez-vous 15 minutes pour un échange ?<br/><br/>Je peux vous envoyer une courte fiche ou une démonstration de cinq minutes si cela vous intéresse.<br/><br/>Bien cordialement,<br/>[Prénom Nom]”*

## English version

*“Subject: IASD research prototype - technology Operational Risk investigation<br/><br/>Hello [Name],<br/><br/>I am completing a research prototype as part of my applied mathematics and IASD studies. RiskSignal explores a focused question: can a single workspace help an analyst connect incidents, indicators, controls and supplier dependencies before deciding what to examine next? The demonstration uses synthetic data and an illustrative loss model only; it does not predict losses or assess a real bank's risk.<br/><br/>The prototype combines Python, stochastic simulation, dependency analysis, an API and a read-only tool-using assistant. I would value your perspective on the use case and on what would need to be validated before any work with approved data. Would you have 15 minutes for a short conversation?<br/><br/>I can share a one-page overview or a five-minute demo if useful.<br/><br/>Best regards,<br/>[Name]”*

## Why this is a good teaser

- It states the problem and technical substance without attaching the full application or every model detail.
- It makes the synthetic-data and non-prediction boundaries clear up front.
- It asks for a small, specific next step: a 15-minute discussion.
- It offers a one-page summary or short demo only if the recipient wants more.

# 12. Sources and preparation notes

## Project documentation

- README.md: project scope, launch steps, default dataset and demo flow.
- docs/domain_model.md: Operational Risk terms used in the prototype.
- docs/stochastic_model.md and src/risksignal/stochastic/: model definitions and implementation.
- docs/agent_design.md: agent loop, provider boundary, tools, safety limits and traceability.
- docs/limitations.md: synthetic data, model and deployment limitations.
- docs/demo_script.md: current five-minute demonstration route and reference values.
- docs/industry_research.md and docs/research/job_evidence.csv: exploratory job-description review; not a representative survey of the labour market.
- Supplied CV: IASD coursework and quantitative-modelling experience; no contact details are reproduced in this guide.

## Market examples (illustrations only)

The project research notes a full Man Group enterprise-risk analyst description discussing risk/control records, issues, vendor lifecycle, data quality, reporting and workflow improvement. View the listing.
An EY wealth and asset-management risk advisory role discusses RCSAs, controls, automated monitoring and human-led governance. View the role.
These examples illustrate possible responsibilities. They do not prove how all OR teams work, how long tasks take, or that the prototype saves time. The 34-role research sample is exploratory and not representative of the labour market.

> **Final reminder**
> You do not have to sound like an Operational Risk professor. You have to be clear about what you built, what the maths means, what is synthetic, what remains unproven, and what you would test next.

# 13. What the analyst is trying to do

The main priority is to decide what deserves further investigation, why it matters, and which evidence to request next. A simulation is supporting analysis, not the investigation itself.

### 1. Notice a concern

Recurring incidents, an unusual indicator or a supplier problem.

### 2. Investigate

Examine incidents, controls and dependencies to understand what needs checking.

### 3. Use Monte Carlo for the financial question

> *“If incidents become more frequent or costly, what range of annual losses could result under those assumptions?”*

### 4. Decide the follow-up

Request missing evidence and discuss action with the responsible people.

### The link between the investigator and Monte Carlo

The investigator brings together records and can call a Python simulation tool. Monte Carlo adds a conditional financial view: an annual-loss distribution under specified frequency and severity assumptions. The analyst checks the evidence and assumptions before discussing next steps.

**Important:** today, KRIs, control-review status and the dependency graph do not automatically change the loss model. The baseline uses incident counts and positive loss amounts. A stress scenario changes its parameters explicitly. The LLM does not invent those calculations.

> *“RiskSignal helps an analyst investigate a system showing concerning signals, then explore the possible financial consequences through simulation.”*

# 14. Worked example: one payment system

**System:** PAYMENT_ENGINE_01, supporting International Payments.
**Evidence date:** 30 September 2026, not today's date.
**Context:** fictional Atlas Bank; seed-42 synthetic data. The storyline was deliberately constructed. Amounts below are synthetic monetary units, not dollars or euros.

## Step 1 — Notice something worth checking

Select the system. Its latest failed-deployment-rate value is **10.51**, compared with a median of **4.675** over the preceding 12 months. The illustrative median/MAD rule flags an unusual increase (score about **5.39**, above 3.5). This is not a 10.51% probability of an incident, an approved appetite breach or a forecast. Not every latest indicator is flagged.

> *"This indicator prompts a closer look. I now need the incident records and control evidence before interpreting what it means."*

## Step 2 — Investigate the records

In the last 12 months, five recorded incidents total **340,080 units**. Two descriptions mention deployment, two involve Keystone Payment Rails and one concerns a manual change.

| Date | Recorded event | Loss |
|---|---|---|
| 24 Nov 2025 | INC0245: Keystone service degradation affected outbound payments. | 1,635 |
| 16 Feb 2026 | INC0258: failed deployment; rollback after batch errors. | 19,859 |
| 12 May 2026 | INC0277: incorrect manual routing-rule change. | 16,376 |
| 19 Jul 2026 | INC0292: deployment introduced a message-mapping defect; hotfix required. | 161,612 |
| 11 Sep 2026 | INC0306: Keystone API latency delayed submissions. | 140,598 |

## Connect controls and dependencies

**C002 Pre-Release Regression Testing:** review overdue; 420 days since last review. **C004 Vendor Performance Review:** review due; 320 days since last review. **V04 Keystone Payment Rails:** also used by PAYMENT_ENGINE_02.

> *"Deployment-related records and an overdue testing review justify asking for test evidence. They do not prove that C002 failed or caused the incidents. The shared vendor identifies another system to consider, not a proven simultaneous failure."*

# 15. Use Monte Carlo for the financial question

## Step 3 — Ask a specific quantitative question

> *"For PAYMENT_ENGINE_01, compare the baseline with the SHARED_VENDOR_OUTAGE scenario. Explain the assumptions separately from the incident evidence."*

The full five-year history contains **19 incidents**, giving a fitted rate of about **3.80 incidents per year**. A Lognormal severity model is fitted to the 19 positive losses (mu about 10.114; sigma about 1.478, both in log space). The simulation generates **100,000 hypothetical annual totals** using seed 42.

## What happens inside one simulation?

Draw the number of incidents, draw a financial loss for each, then add the losses. **Teaching example only:** one imaginary year could contain two losses of 5,000 and 40,000, totalling 45,000 units. Another could contain no incidents. These numbers explain the mechanism; they are not actual extracted simulation years.

## Baseline versus a deliberately tougher assumption set

The predefined stress assumes **lambda × 1.8**, **mu + 0.2** and **sigma × 1.1**. It therefore models more events and a changed loss distribution. These choices are illustrative, not calibrated to vendor evidence. The graph does not set them.

| Annual-loss measure | Baseline | Illustrative stress |
|---|---|---|
| Mean | 278,433 | 771,375 |
| Median | 163,846 | 498,070 |
| P95 | 896,435 | 2,276,896 |
| ES95 | 1,546,928 | 3,927,230 |

**P95:** the threshold exceeded in roughly 5% of simulated years. **ES95:** the average total in the worst 5%. The stress P95 is about **154% higher** than the baseline P95 under these choices.

> *"This does not say the system will lose 2.28 million next year. It says the selected model and stress assumptions produce that tail threshold. It helps us discuss sensitivity to worse conditions, not establish their probability or cause."*

**Do not confuse the figures:** 340,080 is a recorded synthetic loss total for one 12-month period; 278,433 is a simulated annual average fitted from five years. They describe different things. No control or KRI signal automatically becomes a predicted loss.

# 16. Finish the case with an analyst follow-up

## Step 4 — Decide which evidence to request

| Observed concern | Suggested follow-up, not an automated decision |
|---|---|
| Deployment-related incidents; C002 overdue review | Request release/test records, the latest control review, exceptions and incident root-cause evidence. Ask the relevant owner whether the review was completed or scheduled. |
| V04 incidents; shared supplier | Request vendor incident reports and contingency/recovery evidence. Check relevant PAYMENT_ENGINE_02 records with the responsible people. |
| Unusual latest KRI point | Confirm the definition, input quality, deployment volume and business context. Determine whether the change is meaningful before escalation. |
| Large simulated stress losses | Challenge the scenario multipliers and model assumptions. Discuss whether an expert-designed scenario would better answer the analyst's question. |

## A concise, illustrative analyst summary

> *"The synthetic records justify reviewing release testing and the shared vendor. There are two deployment-related narratives, two vendor incidents and an overdue C002 review. The stress simulation shows that annual-loss outcomes are sensitive to tougher assumptions. I cannot conclude that the control failed, identify a root cause, or forecast losses. I would request test, incident and vendor evidence before discussing further action."*

This paragraph is a **suggested presentation summary**, not a quoted live-LLM result. In the application, read the actual findings, open **Retrieved evidence and tool activity** and check one claim against its source. The mock provider is scripted, not a live language model.

## Longer-term goal: evaluate forecasting with real data

With approved bank data, the research goal can include estimating and testing a **probabilistic annual-loss forecasting model**, with Monte Carlo generating its outcome distribution. That requires meaningful historical records, suitable definitions, parameter and model checks, later-period evaluation and independent review. Monte Carlo itself does not create predictive validity. It would not predict the exact date of the next incident.

An initial real-data pilot would need an institutional partner, permissions and approved security/hosting. No access date is confirmed. First agree one analyst task and reviewed reference cases; assess evidence quality, usefulness, errors, time and cost rather than assume a benefit.

## Your closing sentence

> *"The analyst's priority is to understand what deserves checking and why. The investigator connects the evidence; Monte Carlo supports the financial what-if; a person reviews the assumptions and decides the follow-up."*